

HDOC Day-6 Activity Report

Name: Dhruv Pandey

SAP ID: 590037660

Batch: 15

C Programming - Problem Solving, Algorithms, Test Cases, Compilation and Execution

Question 1 - Cost of Carpeting a Square Room

Problem: A square room has side s meters. Find the carpeting cost when carpet costs r rupees per square meter.

Step 1 - Input, Output and Formula

Input: side (s), carpet rate (r)

Output: area of room and carpeting cost

Formula: Area = $s \times s$; Cost = Area $\times r$

Step 2 - Test Cases

Test Case	Input	Expected Output	Actual Output	Result
1	$s=5, r=7$	Area=25.00; Cost=Rs. 175.00	Area=25.00; Cost=Rs. 175.00	Pass
2	$s=5, r=3$	Area=25.00; Cost=Rs. 75.00	Area=25.00; Cost=Rs. 75.00	Pass
3	$s=9, r=8$	Area=81.00; Cost=Rs. 648.00	Area=81.00; Cost=Rs. 648.00	Pass

Step 3 - Algorithm

1. Start.
2. Read side s and carpet rate r .
3. Calculate area = $s \times s$.
4. Calculate cost = area $\times r$.
5. Display area and carpeting cost.
6. Stop.

Evaluation: The algorithm uses the correct square-area formula and multiplies it by the rate, so it produces the required carpeting cost.

Step 4 - C Program (Screenshot)

```
UW PICO 5.09 File: day6.c

#include <stdio.h>

int main()
{
    float side, rate, area, cost;

    printf("Enter the side of the square room (in meters): ");
    scanf("%f", &side);

    printf("Enter the carpet cost per square meter (in rupees): ");
    scanf("%f", &rate);

    area = side * side;
    cost = area * rate;

    printf("\nArea of the room = %.2f square meters\n", area);
    printf("Cost of carpeting = Rs. %.2f\n", cost);

    return 0;
}
```

Step 5 and 6 - Compilation, Execution and Evaluation

The program was compiled using clang and executed for the prepared test cases. The actual outputs match the expected outputs; therefore, all three test cases pass.

```
dhruvpandey@DHRUVs-MacBook-Air ~ % clang day6.c -o day6
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day6
Enter the side of the square room (in meters): 5
Enter the carpet cost per square meter (in rupees): 7

Area of the room = 25.00 square meters
Cost of carpeting = Rs. 175.00
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day6
Enter the side of the square room (in meters): 5
Enter the carpet cost per square meter (in rupees): 3

Area of the room = 25.00 square meters
Cost of carpeting = Rs. 75.00
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day6
Enter the side of the square room (in meters): 9
Enter the carpet cost per square meter (in rupees): 8

Area of the room = 81.00 square meters
Cost of carpeting = Rs. 648.00
dhruvpandey@DHRUVs-MacBook-Air ~ %
```

Question 2 - Painting Cost and Border Length of a Parallelogram

Problem: Find the painting cost and border length of a parallelogram-shaped board using base, height, adjacent side and painting rate.

Step 1 - Input, Output and Formula

Input: base (b), height (h), adjacent side (a), painting rate (r)

Output: area, painting cost and border length

Formula: Area = $b \times h$; Painting Cost = Area $\times$ r; Border Length = $2 \times (b + a)$

Step 2 - Test Cases

Test Case	Input	Expected Output	Actual Output	Result
1	b=1,h=243,a=54,r=2	Area=243.00; Cost=Rs. 486.00; Border=110.00 m	Area=243.00; Cost=Rs. 486.00; Border=110.00 m	Pass
2	b=5,h=6778,a=44,r=3	Area=33890.00; Cost=Rs. 101670.00; Border=98.00 m	Area=33890.00; Cost=Rs. 101670.00; Border=98.00 m	Pass
3	b=78,h=55,a=42,r=55	Area=4290.00; Cost=Rs. 235950.00; Border=240.00 m	Area=4290.00; Cost=Rs. 235950.00; Border=240.00 m	Pass

Step 3 - Algorithm

1. Start.
2. Read base, height, adjacent side and painting rate.
3. Calculate area = base $\times$ height.
4. Calculate painting cost = area $\times$ rate.
5. Calculate border length = $2 \times$ (base + adjacent side).
6. Display area, painting cost and border length.
7. Stop.

Evaluation: The algorithm correctly uses the parallelogram area and perimeter formulas and computes the painting cost from the area.

Step 4 - C Program (Screenshot)

```
UW PICO 5.09 File: day6.2.c
#include <stdio.h>

int main()
{
    float base, height, side, rate;
    float area, paintingCost, borderLength;

    printf("Enter the base of the parallelogram (in meters): ");
    scanf("%f", &base);

    printf("Enter the height of the parallelogram (in meters): ");
    scanf("%f", &height);

    printf("Enter the adjacent side of the parallelogram (in meters): ");
    scanf("%f", &side);

    printf("Enter the painting rate per square meter (in rupees): ");
    scanf("%f", &rate);

    area = base * height;
    paintingCost = area * rate;
    borderLength = 2 * (base + side);

    printf("\nArea of the parallelogram = %.2f square meters\n", area);
    printf("Painting cost = Rs. %.2f\n", paintingCost);
    printf("Border length = %.2f meters\n", borderLength);

    return 0;
}
```

Step 5 and 6 - Compilation, Execution and Evaluation

The program was compiled and executed for three test cases. The displayed area, painting cost and border length match the expected results, so all test cases pass.

```
dhruvpandey@DHRRUVs-MacBook-Air ~ % clang day6.2.c -o day6
dhruvpandey@DHRRUVs-MacBook-Air ~ % ./day6
Enter the base of the parallelogram (in meters): 1
Enter the height of the parallelogram (in meters): 243
Enter the adjacent side of the parallelogram (in meters): 54
Enter the painting rate per square meter (in rupees): 2

Area of the parallelogram = 243.00 square meters
Painting cost = Rs. 486.00
Border length = 118.00 meters
dhruvpandey@DHRRUVs-MacBook-Air ~ % ./day6.2
Enter the base of the parallelogram (in meters): 5
Enter the height of the parallelogram (in meters): 6778
Enter the adjacent side of the parallelogram (in meters): 44
Enter the painting rate per square meter (in rupees): 3

Area of the parallelogram = 33890.00 square meters
Painting cost = Rs. 101670.00
Border length = 98.00 meters
dhruvpandey@DHRRUVs-MacBook-Air ~ % ./day6.2
Enter the base of the parallelogram (in meters): 78
Enter the height of the parallelogram (in meters): 55
Enter the adjacent side of the parallelogram (in meters): 42
Enter the painting rate per square meter (in rupees): 55

Area of the parallelogram = 4290.00 square meters
Painting cost = Rs. 235950.00
Border length = 248.00 meters
dhruvpandey@DHRRUVs-MacBook-Air ~ %
```